Development verification record — 2026-10-17

These checks establish implementation behavior. Comparative sample quality is assessed separately in the artifact-derived paper and paired analysis. This record retains the initial development checks and subsequent release verification.

Sequence and source provenance

The public repository and pre-specified protocol were pushed in commit `55dc2e4`. Core numerical/training implementation (`2c9a45d`) and CLI (`a3dd622`) were pushed before the first MNIST optimizer run. The first GPU run used `a3dd622`. Subsequent evaluation/deadline support was pushed in `5b69c90`; the fixed-batch/pilot scheduler was pushed in `b0eb1f5`. Full reporting/export support was pushed in `af53767`. Git history and run `source_revisions` retain the exact full revisions.

The originating user instructions and implementation plan specified local training, a 24 GPU-hour ceiling, public code/checkpoints, and exploratory paired inference. No runtime LLM was introduced. The implementation used TorchCFM, torchdiffeq, torch-fidelity, PyTorch, POT, and SciPy instead of recreating their maintained algorithms. AKOrN source had no repository license during inspection, so its published equations were implemented independently.

Experiments and observations

Check	Observed result	Interpretation
MNIST GPU smoke, 50 updates	Completed; about 17.68 training seconds; peak allocated memory 203,779,584 bytes	CUDA forward/backward/optimizer/checkpoint path executes
Resume to 100 updates	Completed; cumulative training time 33.96 seconds; EMA validation velocity MSE 2.00869	Saved run state can continue on GPU; early EMA remains close to initialization
Exact CPU resume test	Resumed and uninterrupted model/EMA/noise-generator states match exactly	Optimizer-boundary persistence is complete in the tested CPU setting
Fixed batch, 500 updates	MSE 2.255753 → 0.000602648; ratio 0.000267161	Full oscillator model can learn fixed production-path velocity targets
Same-input determinism	Exact repeated predictions in the fixed-batch GPU check	No hidden state is carried across these forward calls
Geometry	Maximum norm error 5.96e-8; maximum tangent inner-product magnitude 3.05e-7	FP32 spherical operations meet the predeclared 1e-5 numerical tolerance
CPU sampling	16 samples at 32 NFE completed; full float outputs and grid saved	GPU is not required for inference
GPU sampling/evaluation smoke	128 samples at 32 NFE completed; MNIST pixel SWD 1.101009	Full evaluation path works; this noise-like early model is not a quality result
Portable checkpoint reload	Original and safetensors-export CPU samples exactly equal; maximum difference zero	Safe export preserves the actual inference function
Budget enforcement	Isolated deliberately blocked worker was terminated; reservation retained	The ceiling is enforced independently of cooperative model code

Check	Observed result	Interpretation
Report/PDF workflow	Markdown and PDF records generated; first page and equations inspected	Pandoc/Typst artifacts exist and render the mathematical method
Planning-power reproduction	0.2406 at standardized effect 0.8 using 20,000 Gaussian simulations	Reproduces the pre-training low-power warning; not observed empirical power

The CPU preview after 100 optimizer updates visually remained noise-like. This was expected from a very short development run and a slowly updated EMA; it was retained rather than selected away. A longer fixed-time pilot, not an extension based on significance, was already part of the protocol.

Issues, decisions, and limits

The initial resume invocation omitted the required path after `--resume`; argparse rejected it before computation. The corrected documented form is `ofm train --resume <run-directory>`. Import-order lint findings were repaired while implementation progressed; CI run 38977501341 failed only an import-format check, repaired in `549327f`. A third-party POT SyntaxWarning did not affect numerical tests. No nonfinite-loss or gradient failure was observed in the checks above. A CLI help traceback was removed, and the scheduler was hardened to load actual frozen pilot settings instead of allowing later environment-default drift.

Tests for missing flow/training/evaluation/reporting/publication interfaces were introduced before those implementations, and their initial collection failures were observed. Some initial model tests could only execute after CUDA dependencies finished installing; this record does not claim every line followed a strict observed red-green sequence. The initial complete local suite contained 27 passing tests; additional help/freeze regression tests were added for the two later repairs. CI must pass on the merged implementation.

The 20-minute-per-architecture MNIST pilot uses the pre-specified optimizer settings and validates generalization/visual behavior before freezing CIFAR settings. No CIFAR result was used to choose these parameters. A fixed per-update EMA decay and differing architecture throughput are potential comparative confounders; they are reported as limitations rather than silently adjusted after outcomes. GPU utilization and system background load are not hardware-independent compute measures.

Completed pilot and metric preflight

All three MNIST pilots completed their original 20-minute budgets. Their fixed first-64 sample grids were inspected: they contain recognizable but rough digits, including ambiguous shapes. Full measurements and images are retained in the generated experiment records rather than repeated here. The freeze was written at `20261016T233806.675239Z`, source `7926ff79c12be5d85eadb49b81aad5d3acfaedcd`, with an empty changes list. CIFAR scheduling inherits these actual saved settings. The CIFAR decoded-pixel audit retained 45,000 training, 5,000 validation, and 10,000 test examples, with no exact cross-split overlap found; this is not a near-duplicate audit.

Before any CIFAR optimization, an untrained oscillator generated 128 validation-check samples. The real torch-fidelity feature/KID/FID path completed, and repeating all metrics from the saved tensors yielded a maximum absolute difference of zero. This preflight used 128-

image KID subsets and validation data, so it is not a primary 5,000-image test comparison. Its artifact group is `runs/checks/cifar10-metric-smoke`.

The publicly downloaded trained oscillator pilot also generated 64 images on CPU at 32 velocity evaluations. Two fixed CPU input/time pairs gave exactly equal predictions from the public safetensors weights and the original EMA checkpoint. The saved public-weight SHA-256 is `06201135c2255dcc8b2ee988f9ce128ce6fe2e6eb35f3fd142b3a4db415fec96`; the checkpoint identity and all preview tensors are retained under the pilot's `cpu-public-check` directory. This checks a published trained artifact, in addition to the earlier development export test.

The implementation review and expanded regression suite concluded with 33 passing tests and a passing Ruff check before the CIFAR schedule. The frozen dependency sync completed before that schedule. These checks justified starting the declared comparison; they did not establish an oscillator quality advantage.

Final training and release verification

All 18 main CIFAR arms and four trained ablations completed their original 45-minute budgets. Their recorded optimization durations range from 2,700.0286 to 2,700.2957 seconds, reflecting the final optimizer-step boundary. Every arm records source revision `c749856d1c9fe07d6ce2f39d01de78f664aea9a8` and the same prepared-data SHA-256, `51ca348f81e4c280d10942cf2fa0d4dd1ce688a16b0088283069b54bbaaf6f97`. The complete experiment index links the individual training records. No scientific model, training, data, sampling, metric, or inference configuration was changed during this comparison. Later source changes concern report composition, PDF layout, and release completeness.

Release inspection found that standalone controls without a checkpoint needed an explicit export path. A regression test first failed when checking the missing real-control partition evidence; the repaired exporter now retains raw controls, failures, associated logs, and full fixed-input verification tensors through an explicit allowlist. It still excludes sibling `.env` and unrelated files. The report summary test also requires all six declared seeds and rejects a more favorable secondary-NFE score as a substitute for the primary observation. The expanded suite has 35 tests; the CI run for the final export repair passed the complete suite and Ruff. The targeted reporting/publication rerun passed all six tests locally.

A subsequent two-seed summary test checks explicit pairing, preservation of a null second-seed effect, rejection of an attractive alternate sampling cost, and incomplete-pair handling. It first failed before the new summary function existed. The expanded CI run passed all 36 tests and Ruff. Its sole warning is a pre-existing invalid escape sequence in a third-party POT docstring, not a numerical or application failure.

Visual PDF inspection exposed two concrete layout defects: long tables overflowed a single page, then numeric columns overlapped after pagination was enabled. Following the Typst table guide, table figures now permit page breaks and table text uses eight-point type. The corrected paper pages and an individual experiment PDF were rendered and inspected; table rows, hashes, paths, and numerical columns were readable. A final display regression caught integer parameter counts being rounded by floating-point formatting. A failing test preceded the one-line repair; counts now print exactly, boolean outcomes remain explicit, and measured floats retain six significant display digits with full precision in their raw records. This brings

the suite to 37 tests. The comparison figure uses the preselected first 64 seed-101 images at 32 evaluations, with the complete fixed grid for every architecture. These repairs affect presentation and evidence retention, not any experimental outcome.

The complete study driver exited with code zero after all 38 scheduled generator evaluations and the real-versus-real repeat control. Both identity interventions exactly reproduced every value of the original 5,000-image tensor. The real-control metric repeat had maximum absolute difference zero. Final accounting was 18.7365855 charged GPU-hours with no active reservation; this includes the earlier development and pilot work. All 22 CIFAR training runs and all scheduled evaluations completed without a recorded experiment failure. The individual control records preserve the full numerical evidence; exact partition indices and generated tensors remain in the artifact release.

The separate read-only progress helper raised `IndexError: list index out of range` while observing the new real-control log before it contained a progress line. The scientific worker continued and completed normally. Direct artifact checks that tolerate an empty log replaced that helper; no experiment was restarted and no budget entry was altered. This auxiliary failure is retained under `runs/checks/progress-monitor-failure`, alongside the earlier CPU cache-initialization failure. Neither is omitted from the generated failure count. The complete final study logs contain no `NaN`, `Infinity`, traceback, error, or warning entries.

Fixed first-64 grids were inspected for the main architectures, trained removals, state shuffling, and the untrained oscillator. The trained images remain rough; no-coupling samples show conspicuous patch discontinuities and the untrained grid is noise-like. These are visual observations of the fixed grids, not additional quantitative quality claims or a causal identification of synchronization. Timing remains descriptive of the live workstation: occasional CPU tests and report preparation overlapped evaluation, and background load was not experimentally isolated.

All 31 reported checkpoints were staged through the production publisher and reloaded on CPU from their safetensors exports. Every state tensor and resolved configuration matched the original checkpoint, and both fixed input/time predictions per model matched exactly, with maximum absolute difference zero. This includes five MNIST and 26 CIFAR checkpoints across trained, untrained, and diagnostic roles. The complete reload record identifies every original checkpoint and exported-weight hash. Inputs, times, and both prediction tensors are preserved in `runs/checks/final-export-reloads/fixed_batch.safetensors`, SHA-256 `050793bd048da810ce6135f14dd5d23d4328f0b3488d8b7ecbb18a80ec123d43`. CPU verification used four threads and deterministic algorithms; it does not claim cross-device bitwise equivalence.

Sources: protocol, implementation contracts, Karpathy's training recipe, PyTorch serialization, SciPy permutation tests.